

Project Title : Intelligent Multi-Sensor Smart Windshield Monitoring System with UWB Actuator Control and LoRa-Based V2V Hazard Communication

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ABSTRACT:

This project presents an intelligent multi-sensor smart windshield monitoring system designed to improve driver safety under poor visibility conditions caused by rain, fog, and sudden windshield obstruction. The system uses an ESP32 microcontroller integrated with rain, humidity, and infrared sensors to continuously monitor environmental conditions in real time. Based on the sensor inputs, the system automatically activates the windshield wiper and defogging mechanisms to enhance visibility and reduce driver workload. Ultra-Wideband (UWB) communication is implemented for low-latency wireless control between the sensing and actuator units inside the vehicle, while LoRa communication is used to transmit hazard alerts to nearby vehicles under low-visibility conditions. An ESP32-CAM module is further incorporated to provide real-time visual assistance during windshield obstruction. The developed prototype demonstrates reliable operation, fast response time, and effective wireless communication, making it a compact and scalable solution for intelligent automotive safety applications.

Objectives	• To detect rain, fog, and windshield obstruction using sensors • To automate wiper and defogger operation using ESP32 • To implement UWB-based wireless actuator control • To implement LoRa-based V2V hazard communication • To improve driver safety and reduce reaction time
Knowledge acquired from the courses studied	The development of this project involved the application of knowledge gained from courses related to embedded systems, microcontrollers, wireless communication, IoT, and automotive electronics. Concepts such as sensor

	interfacing, real-time data acquisition, relay and actuator control, and communication protocols including UART, I2C, and SPI were utilized during implementation. The project also provided practical experience in ESP32 programming using Arduino IDE, wireless communication using UWB and LoRa technologies, system integration, hardware debugging, and real-time monitoring and control for automotive safety applications.
Realistic Constraints	The development of the proposed system is subject to several practical constraints related to hardware, communication, and environmental conditions. The project is implemented using prototype-level embedded hardware with limited processing capability and memory resources. Sensor performance may vary due to environmental noise, calibration inaccuracies, and changing weather conditions. Communication range and reliability of UWB and LoRa modules can also be affected by interference and physical obstacles. In addition, the current system is tested under controlled laboratory conditions and does not include full-scale automotive integration or automotive-grade components. Power management, compact hardware packaging, and real-time video processing are also limited by the capabilities of the selected embedded platforms.
Standards to be referred/followed	The proposed system follows relevant embedded system and wireless communication standards to ensure reliable operation and interoperability between different modules. UART, I2C, and SPI protocols are used for serial communication and sensor interfacing within the embedded platform. IEEE 802.15.4a standards related to Ultra-Wideband (UWB) communication are considered for low-latency and short-range wireless data transmission between the sensing and actuator units. LoRa communication is implemented based on Low Power Wide Area Network (LPWAN) principles for long-range and low-power vehicle-to-vehicle hazard communication. IEEE 802.11 standards are utilized through the ESP32-CAM module for Wi-Fi-based real-time video streaming and visual assistance. In addition, standard embedded system design practices and ESP32 hardware interfacing guidelines are followed to ensure stable and efficient system performance.
Multidisciplinary tasks involved	The proposed project involves multidisciplinary tasks combining concepts from embedded systems, wireless communication, automotive electronics, and IoT-based automation. The system integrates sensor interfacing, real-time data processing, actuator control, and wireless communication using UWB and LoRa technologies. Knowledge of microcontroller programming, communication protocols, and hardware integration is required for implementing the sensing and actuator modules. The project also involves automotive safety concepts, real-time monitoring, and human-machine interaction through display interfaces and camera-based assistance. In addition, system testing, debugging, and prototype development are carried out to ensure reliable operation under different environmental conditions.
Deliverables/ Outcomes	The proposed project delivers a functional prototype of an intelligent smart windshield monitoring system capable of detecting rain, fog, and windshield obstruction in real time. The system successfully automates wiper and defogger operations using sensor-based control and demonstrates reliable actuator response through UWB-based wireless communication between the sensing and actuator nodes. LoRa-based vehicle-to-vehicle communication is implemented to transmit hazard alerts under low-visibility conditions, improving driver awareness and safety. The project also provides real-time visual assistance using an ESP32-CAM module during windshield obstruction scenarios. Overall, the developed system demonstrates a compact, low-cost, scalable, and energy-efficient solution for automotive safety applications.

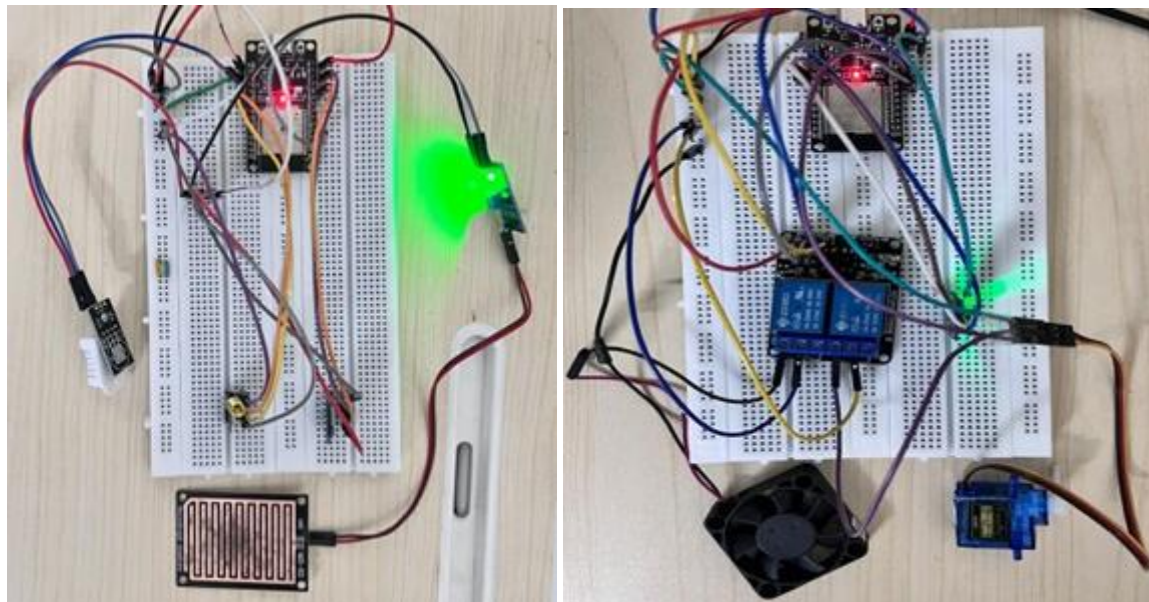
SUSTAINABLE DEVELOPMENT GOALS:

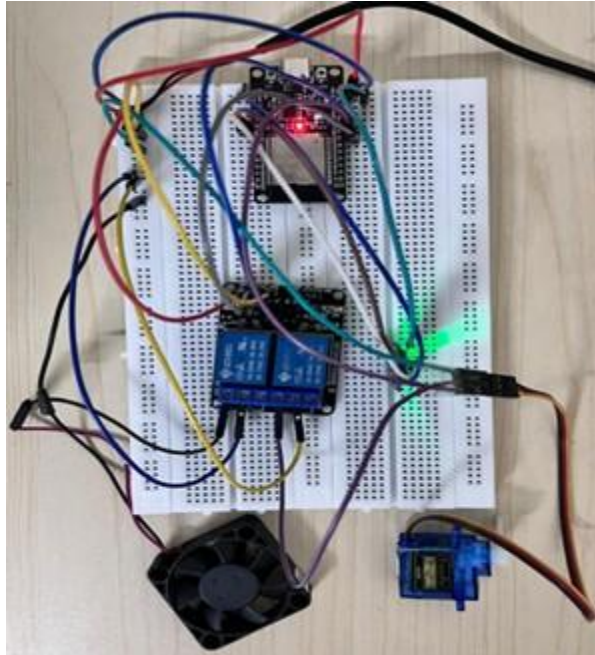
By advancing road safety and contributing to smarter transportation infrastructure, the pro-

posed framework aligns with several United Nations SDGs, namely SDG 3 (Good Health and Well-being), SDG 9 (Industry, Innovation, and Infrastructure), and SDG 11 (Sustainable Cities and Communities).



Final prototype picture :





References:

1. H. Guo, Y. Wang, and Z. Liu, "Research on Automotive Safety Based on Multi-Sensor Information Fusion Technology," *Journal of Physics: Conference Series*, vol. 1885, no. 4, pp. 1-7, 2021.
2. M. Centenaro, L. Vangelista, A. Zanella, and M. Zorzi, "A LoRa-Based Reliable and Low-Power Vehicle-to-Everything (V2X) Communication Architecture," *IEEE Internet of Things Journal*, vol. 6, no. 3, pp. 1-10, 2019.
3. S. Karthikeyan and R. Priyadarshini, "Intelligent Windshield for Automotive Vehicles," *International Journal of Engineering Research & Technology (IJERT)*, vol. 9, no. 6, pp. 1-5, 2020.